

PLUG PULLER · TWO-SIDED PULLER

Quick start, two-sided puller

The shortest path from a stuck plug to a printed two-sided puller that fits it and your hand: get the file, measure, fill in the four Customizer steps, print, assemble, use.

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Which tool this is

This guide is for the two-sided puller, `src/Plug_Puller_Two_Sided.scad`: two serrated plates that zip-tie around the plug and close across it, for a plug 24 mm thick or more and for a plug held from both sides, a USB-C tip or a round extension-cord plug. A thinner wall plug is the one-sided puller's job, and that tool has its own guide.

The Customizer's four steps are your plug, your size, the attachment and the print layout. This quick start covers getting the file, measuring your plug and your hand or matching a card, each step's dials with a picture, then printing, assembly and use.

If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

How to read the pictures

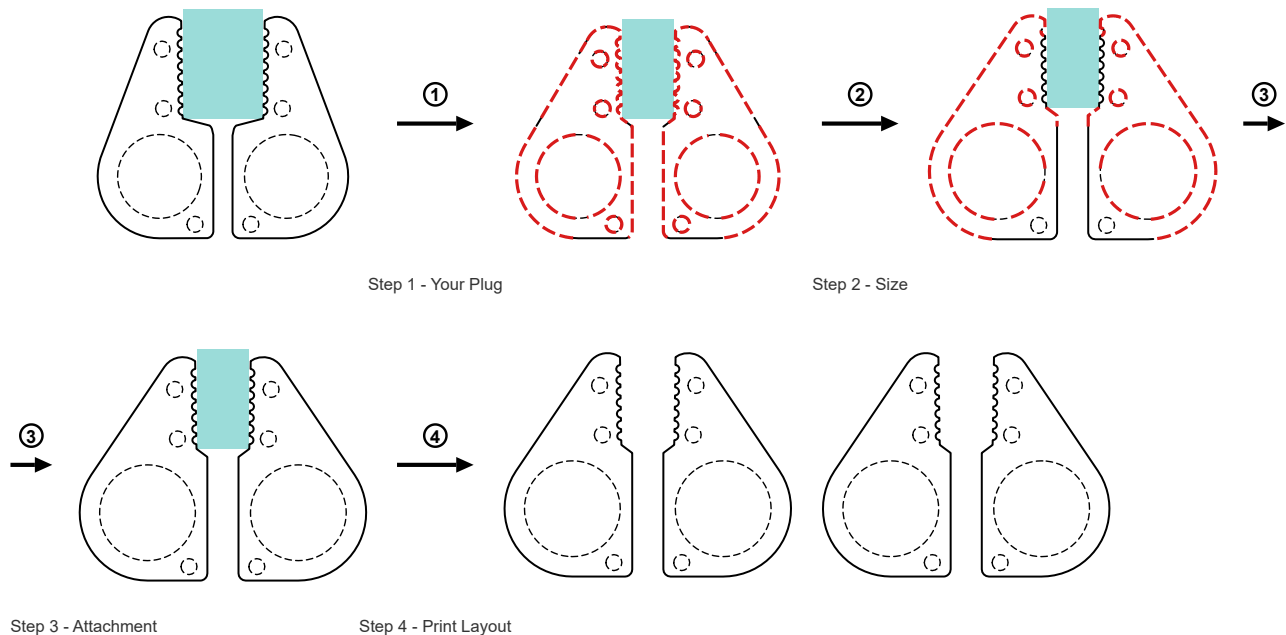
— Black: the tool

Teal: your plug

- - - Red dashed: the edges this dial moved

①

the numbers match the key beside the picture



Five stages of the two-sided puller, left to right, the top row first, the plug end at the top; red dashes mark the edges each step moved. 1, the defaults: the tool as the file opens, with a 20 mm wide plug in teal. 2, Step 1 - Your Plug: plug length 23 mm, plug width at the prong end 13 mm, plug width at the cord end 13 mm, cord thickness 7 mm, plug sides Rounded sides; red on the arms, the finger lobes and the zip stations. 3, Step 2 - Size: hand size Large; red on the arms, the finger lobes and the zip stations. 4, Step 3 - Attachment: attachment Zip ties; no strap slot on a plug this short, so nothing moved. 5, Step 4 - Print Layout: both plates side by side in one file, what you print; nothing marked.

- Step 1: type your plug's numbers and pick Rounded sides or Flat sides
- Step 2: pick your hand size
- Step 3: pick how it attaches; a plug this short gets no strap slot
- Step 4: both plates in one file

Get the file

The two-sided puller is the file `src/Plug_Puller_Two_Sided.scad`. There are two ways to open it and fill in its form: in your web browser with nothing to install, or in the free OpenSCAD program on your computer. Both show the same form, called the Customizer.

In your browser

The OpenSCAD Playground runs the whole Customizer in your browser, on a laptop or a phone. Nothing is uploaded: your numbers stay on your device.

1. Open [the two-sided puller in the OpenSCAD Playground](#). The first visit downloads the OpenSCAD engine (10 to 20 MB), then the model. You will see code on the left (ignore it) and a preview on the right.
2. Open the **Customize** panel: on a wide screen it is a panel or a tab beside the editor, on a phone a tab at the bottom.

If the link does not load the model, download the whole tool as one file, `dist/Plug_Puller_Two_Sided_SingleFile.scad`, and drag it into <https://ochafik.com/openscad2/>.

On your computer

OpenSCAD is the free program that turns your numbers into a printable file. You use one panel of it and never touch the code.

1. Download OpenSCAD from <https://openscad.org/downloads.html>: the **Development Snapshot** for your system. This project is tested with the snapshot of 2026-01-03; any recent snapshot works. The regular release works too, only slower.
2. Get the project: on its GitHub page press the green **Code** button, then **Download ZIP**, and unzip it anywhere. Or download only `dist/Plug_Puller_Two_Sided_SingleFile.scad`, the whole tool in one file.
3. Open `src/Plug_Puller_Two_Sided.scad` (or the single file) in OpenSCAD: double-click it, or use **File** ▶ **Open**. A wall of code appears in an editor pane. Ignore it; you will not touch it.
4. Show the Customizer, the form you type into: in the **View** menu, make sure **Hide Customizer** is unchecked. In older versions it is **Window** ▶ **Customizer**.

The form's sections read top to bottom in the order you decide things: **Step 1 - Your Plug**, **Step 2 - Size**, **Step 3 - Attachment**, then **{step4}**. Everything below Step 4 is optional.

Measure your plug and your hand

Everything here is in mm: a US plug is about 25 mm wide, so a 1 on your paper means you measured in inches. You need a caliper or a ruler with mm marks, the plug in its outlet, and your own hand only if you pick Measure my hand. Print the measuring form at 100 % and fill it in as you go: the sections below are the form's rows, in the same order,

and the card names (R1, C1, F1 / F2) are the cards of the printed measuring stencil, described under Match a card instead of measuring.

The two-sided puller's widths are the size the two plates close across, so measure across the plug the way the plates will grip it. The thickness pair, the wall plate style and the hand width belong to the one-sided puller only.

Plug length

Customizer name: **measure_plug_length**

Form row 2

With the plug in the outlet, ruler from the wall plate face to the plug's back face. On a laptop or charger plug, measure from the device's edge.

Typical	20 to 50 mm
Example	23 mm
Stencil	card R1

Plug width at the prong end

Customizer name: **measure_plug_width_prong_end**

Form row 3

Caliper across the plug body just behind the prongs, the size the two plates close across. On a USB-C or charger tip, just behind the metal tip; on a round plug it is the diameter.

Typical	12 to 40 mm
Example	13 mm
Stencil	card R1

Plug width at the cord end

Customizer name: **measure_plug_width_cord_end**

Form row 4

The same direction across the plates, measured where the cord leaves the plug body. Skip the soft rubber strain relief.

Typical	12 to 40 mm
Example	13 mm
Stencil	card R1

Cord thickness

Customizer name: **measure_cord_thickness**

Form row 5

Caliper across the cord just behind the plug, on its thin side: flat cord the narrow way, round cord the diameter. With the stencil, the smallest C1 slot that slips over the cord is this number.

Typical	3 to 9 mm
Example	7 mm
Stencil	card C1

Plug sides

Customizer name: **plug_sides**

Form row 6

Look at the plug body where the plates will grip it. Rounded sides for a round cord plug, a USB-C or charger tip; Flat sides for a boxy plug.

Choices	Rounded sides, Flat sides
----------------	---------------------------

Finger width

Customizer name: **measure_finger_width**

Form row 9

Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole. With the stencil, find the smallest F1 or F2 hole your finger passes through comfortably and subtract 5 from its number.

Typical	16 to 24 mm
Example	22 mm
Stencil	card F1 / F2

No caliper? Take a ring that fits that finger snugly, measure the ring's inner diameter in mm and add 1.5 mm.

Strap width

Customizer name: **strap_width**

Form row 11

Read the width on the strap's packaging. ONE-WRAP comes in 10, 13, 16, 20 and 25 mm.

Typical	10 to 25 mm
Example	15 mm

Sanity check: each plug width is a two-digit number, roughly 12 to 45, and the prong-end width is usually the bigger one; the finger width is roughly 14 to 32. A number like 1.3 is inches: measure again with the mm side.

When you measure for someone else, a relative or a client, measure their hand for the finger and hand rows and their outlet and plug for the plug rows. Doubtful between two values? Round up: a slightly roomy fit works, a tight one

does not.

Match a card instead of measuring

The measuring stencil is a set of thin printed cards. Each has a two-letter name raised on its face, so you can find it by touch. Hold your plug in the plug cards: if it fills one card's openings, snug with no big gaps, that card names the preset to pick in Step 1, and you type no numbers at all.

Card	What it is	What it answers
P1	The flat 2-prong lamp plug (NEMA 1-15)	the preset Flat 2-prong lamp plug - NEMA 1-15
P2	The standard 3-prong plug (NEMA 5-15)	the preset Standard 3-prong plug - NEMA 5-15
P3	The heavy-duty extension cord plug (NEMA 5-15)	the two-sided puller's preset Heavy-duty extension cord - NEMA 5-15
P4	The wide 2-prong appliance plug (NEMA 1-15)	the one-sided puller's preset Wide 2-prong appliance plug - NEMA 1-15
R1	A ruler: raised mm ticks, numbers every 10 mm, and a notch every 10 mm you can count by touch	the plug's length, widths and thicknesses
C1	A cord gauge: open slots 3 to 9 mm wide along one edge, slid onto the cord from the side	the cord thickness: the smallest slot that slips over the cord
F1, F2	Finger holes 15 to 25 mm across (F1) and 26 to 32 mm (F2)	your finger width: the smallest hole your middle finger passes through comfortably, minus 5

Each P card has three openings: W for the plug's width, T for its thickness, and an open slot that slides onto the cord. If no card fits, your plug is between presets: measure it instead. The measured tool fits better than any preset.

Printing the cards. Print `stl/Measuring-Stencil/Visual/Measuring-Stencil_Visual_All-Cards.stl`: 1.2 mm thick cards, no supports, any rigid filament. Or print one card at a time from `stl/Measuring-Stencil/`. The cards pack onto sheets for a 200 by 200 mm bed. For a smaller bed, open `Measuring_Stencil.scad` in OpenSCAD, set `bed_width` and `bed_depth` to your bed, and export `part_index` 1, then 2, and so on, one sheet at a time. `part_index` 0 previews every sheet at once; do not print that one.

The tactile version (`stl/Measuring-Stencil/Tactile/`, or `label_mode` set to `Tactile` in the file) prints every label as a raised character at ADA size and adds a Grade 2 braille title flap to every card. The flap prints leaning back, held by thin fins. Snap the fins off, then fold the flap away from the card until it lies flat, so the braille lands face up beyond the card's edge. Fold once, gently; a PETG or PP hinge folds more reliably than PLA.

No 3D printer yet? The last pages of the printed quick start are paper versions of the cards at true size: the plug outlines, a 100 mm ruler and the finger circles. Print them at 100 % (actual size, never fit to page) and check the 50 by 50 mm square with a ruler before you trust them. On their own they are `stencil-sheet.svg` and `stencil-sheet-2.svg`.

The measuring form

The form is printed at true size on page 00, at the end of this guide; the paper stencil sheets follow it. Print the form at 100 % (actual size, never fit to page) and check its 50 mm bar with a ruler before you trust it. Fill in the blanks top to bottom as you measure, then type the numbers into the Customizer in the same order.

#	Customizer name	What you measure	Default	Yours
1	plug_preset	Plug preset: pick one	Measure my plug	_____
2	measure_plug_length	With the plug in the outlet, ruler from the wall plate face to the plug's back face.	25.5 mm	_____
3	measure_plug_width_prong_end	Caliper across the plug body just behind the prongs, the size the two plates close across.	20 mm	_____
4	measure_plug_width_cord_end	The same direction across the plates, measured where the cord leaves the plug body.	20 mm	_____
5	measure_cord_thickness	Caliper across the cord just behind the plug, on its thin side: flat cord the narrow way, round cord the diameter.	4 mm	_____
6	plug_sides	Look at the plug body where the plates will grip it.	Rounded sides	_____
7	show_plug_preview	Show the see-through plug: leave on	on	_____
8	size	Hand size: pick one	Medium	_____
9	measure_finger_width	Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole.	20 mm	_____
10	attachment	Attachment: pick one	Zip ties + Velcro strap	_____
11	strap_width	Read the width on the strap's packaging.	15 mm	_____
12	print_layout	Print layout: pick one	Both plates	_____

What the sheet shows. The left column is the numbered list above, one row per dial in the Customizer's Step order, each with the dial's name, its plain title and either a blank after the default or a tick box per choice with the default marked. The right half is a schematic plug in teal drawn from the two-sided puller's defaults at 1.5 to 1, a top view with the prong end at the left against a wall plate line, the cord's end as a small circle, a bar of four knuckles at 1:1 and a strap bar at 1:1. A straight black arrow runs from each measured row's blank to a red dimension line on the schematic, the row's number in a red circle at the line:

- Arrow 2 points to the plug's length on the top view, from the wall plate to the plug's back end.
- Arrow 3 points to the plug's width at the prong end, the left edge of the top view.
- Arrow 4 points to the plug's width at the cord end, the right edge of the top view where the cord leaves.
- Arrow 5 points to the cord's thickness on the small circle, the cord seen end on.
- Arrow 9 points to one knuckle of the knuckle bar, drawn 20 mm wide at 1:1.
- Arrow 11 points to the strap bar's width, a 40 by 15 mm bar at 1:1.
- Row 1, plug preset: 5 boxes, Measure my plug marked as the default; the choices are Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15.
- Row 6, plug sides: 2 boxes, Rounded sides marked as the default; the choices are Rounded sides, Flat sides.
- Row 7, show the see-through plug: one box marked, leave on.
- Row 8, hand size: 4 boxes, Medium marked as the default; the choices are Small, Medium, Large, Measure my hand.

- Row 10, attachment: 2 boxes, Zip ties + Velcro strap marked as the default; the choices are Zip ties + Velcro strap, Zip ties.
- Row 12, print layout: 2 boxes, Both plates marked as the default; the choices are Both plates, One plate.

Type them into the Customizer in this order.

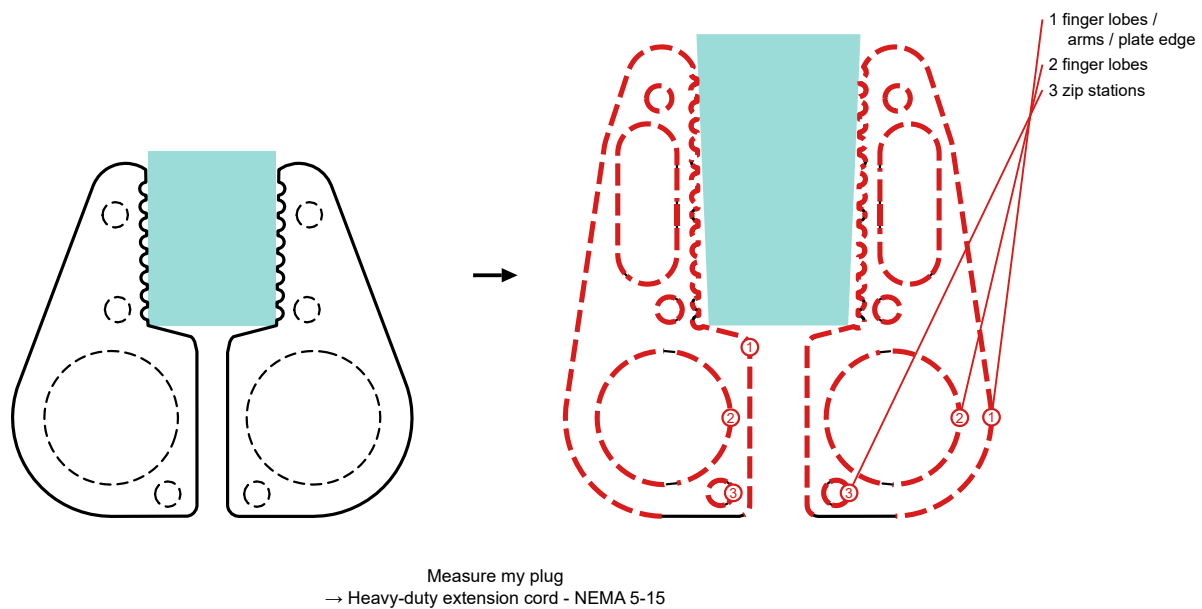
Step 1 - Your Plug

Pick a plug preset, or leave it on Measure my plug and type your plug's numbers; then pick Rounded sides or Flat sides.

Plug preset

Customizer name: **plug_preset**

Fills in the plug's length, both widths and the cord from the measured heavy-duty cord plug, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.



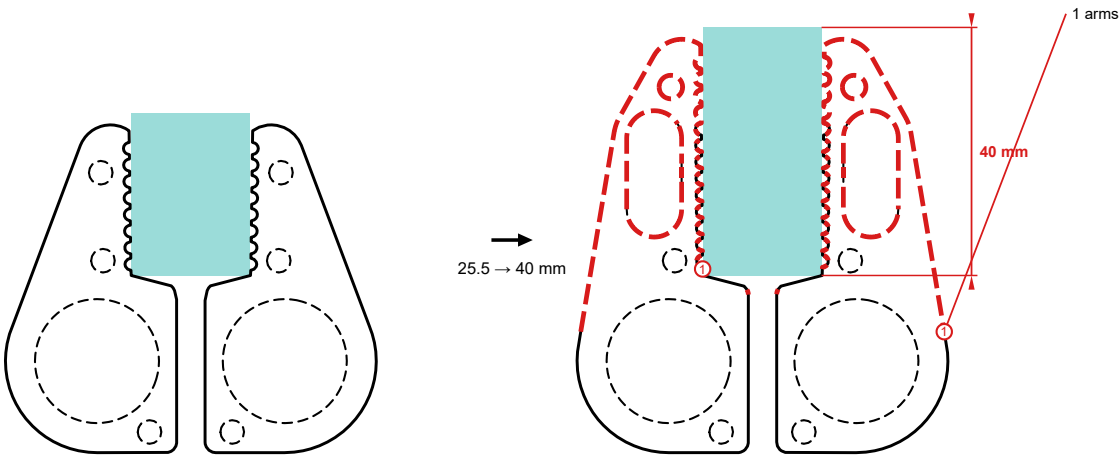
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug preset set to Heavy-duty extension cord - NEMA 5-15. Marked in red: 1, the finger lobes, the arms and the plate edge. 2, the finger lobes: the two rounded lobes in the lower half. 3, the zip stations: the three small holes along each arm.

Default	Measure my plug
Choices	Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15
Moves	arms, finger lobes, plate edge, zip stations

Plug length

Customizer name: **measure_plug_length**

Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.



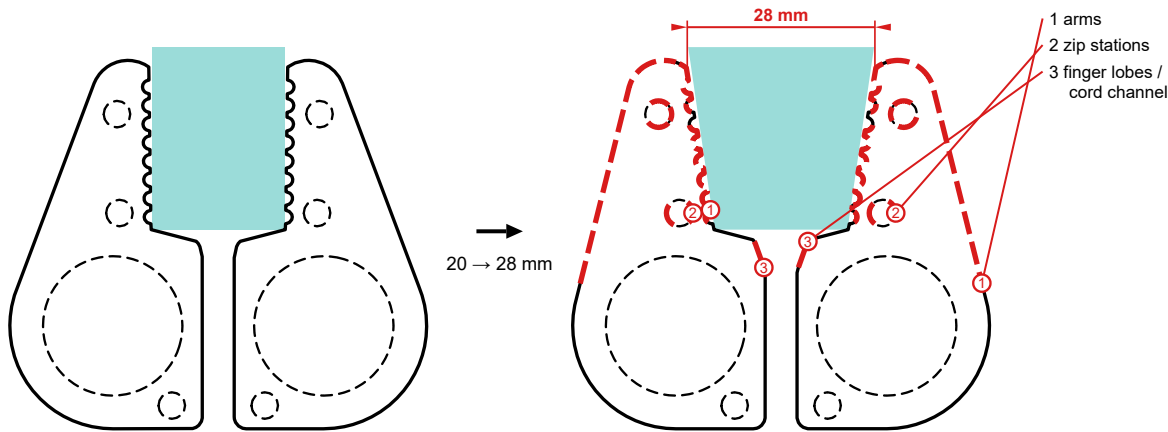
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug length at 40 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. The plate edge, the teeth, the finger lobes and the cord channel stay where they were.

Default	25.5 mm
Range	12 to 85 mm, in steps of 0.5 mm
Moves	arms

Plug width at the prong end

Customizer name: `measure_plug_width_prong_end`

Opens or closes the gap between the arms at their tips, where the plug's prong end sits.



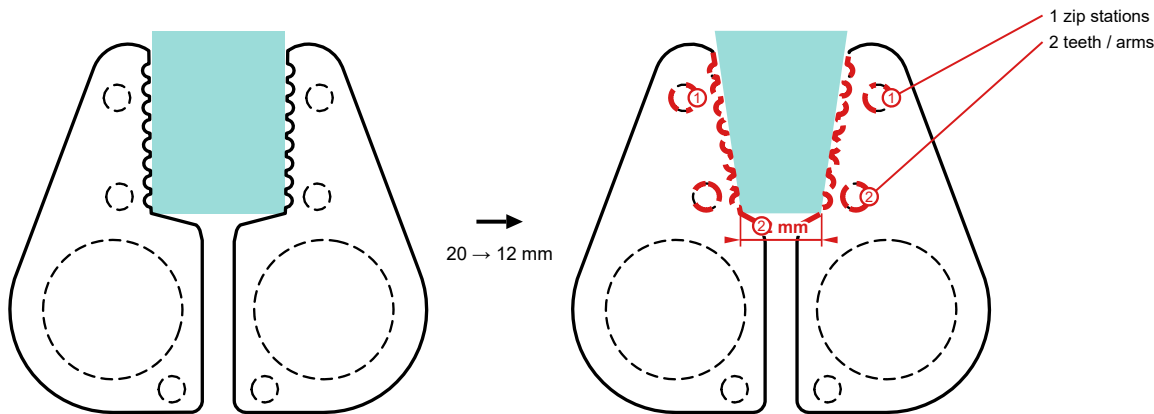
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the prong end at 28 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm. 3, the finger lobes and the cord channel: the two rounded lobes in the lower half; the gap between the lobes at the bottom.

Default	20 mm
Range	5 to 40 mm, in steps of 0.5 mm
Moves	arms, cord channel, finger lobes, zip stations

Plug width at the cord end

Customizer name: `measure_plug_width_cord_end`

Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.



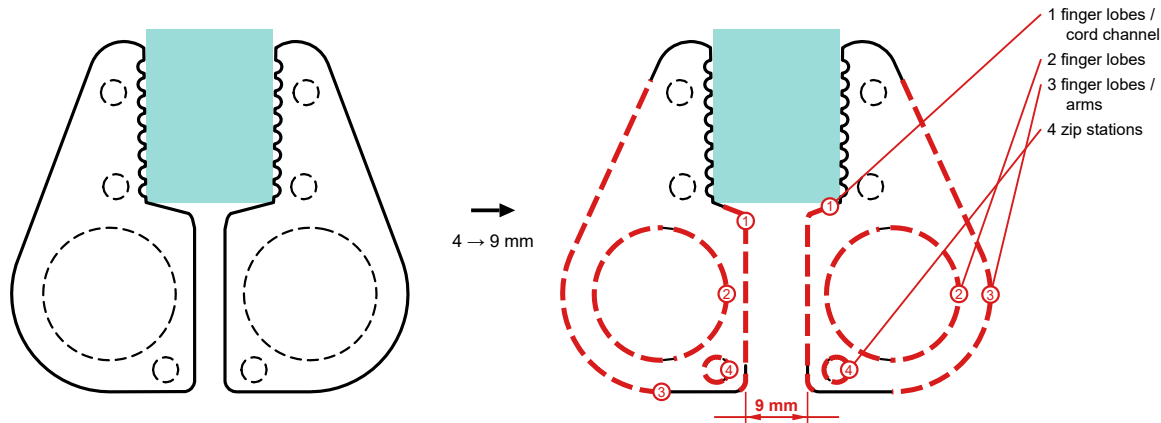
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the cord end at 12 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Default	20 mm
Range	5 to 40 mm, in steps of 0.5 mm
Moves	arms, teeth, zip stations

Cord thickness

Customizer name: **measure_cord_thickness**

Widens the cord channel between the finger holes, and the finger lobes move outward with it.



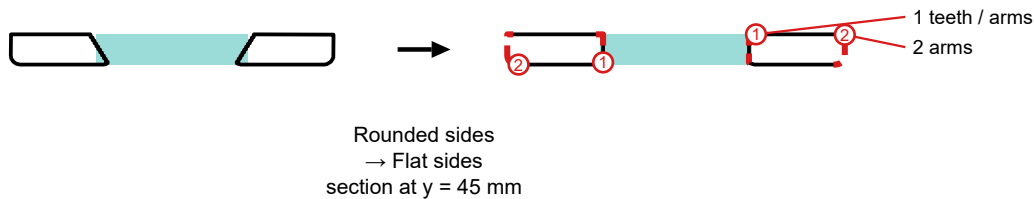
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: cord thickness at 9 mm. Marked in red: 1, the finger lobes and the cord channel. 2, the finger lobes: the two rounded lobes in the lower half. 3, the finger lobes and the arms. 4, the zip stations: the three small holes along each arm.

Default	4 mm
Range	1.5 to 12 mm, in steps of 0.5 mm
Moves	arms, cord channel, finger lobes, zip stations

Plug sides

Customizer name: **plug_sides**

Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.



Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug sides set to Flat sides. Marked in red: 1, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half. 2, the arms: the two toothed arms in the upper half.

Default	Rounded sides
Choices	Rounded sides, Flat sides
Moves	arms, teeth

Show the see-through plug

Customizer name: **show_plug_preview**

Shows or hides the see-through plug in the preview and changes nothing in the printed plates.

No picture. Changes no shape: the see-through plug is a preview aid and is never exported.

Default	On
Choices	On or off (a check box)

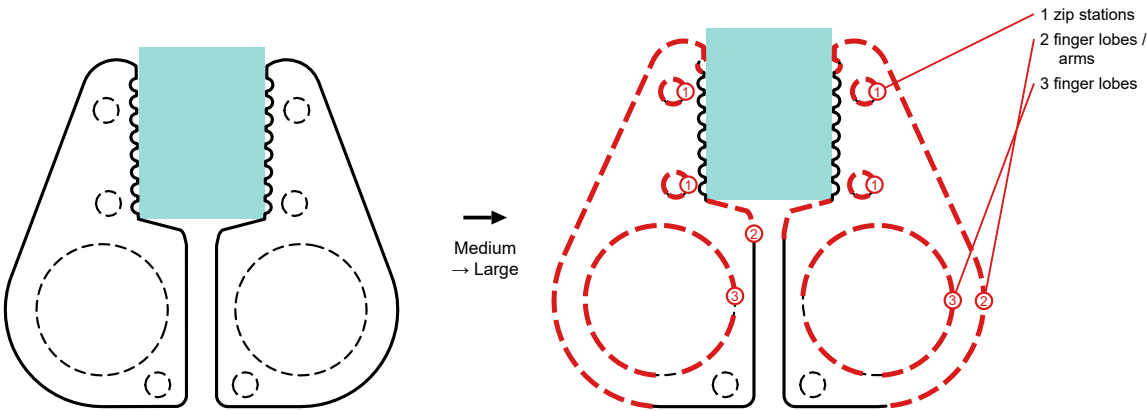
Step 2 - Size

Pick your hand size, or pick Measure my hand and type your finger width.

Hand size

Customizer name: **size**

Widens both finger holes and their lobes, so the whole plate grows around them.



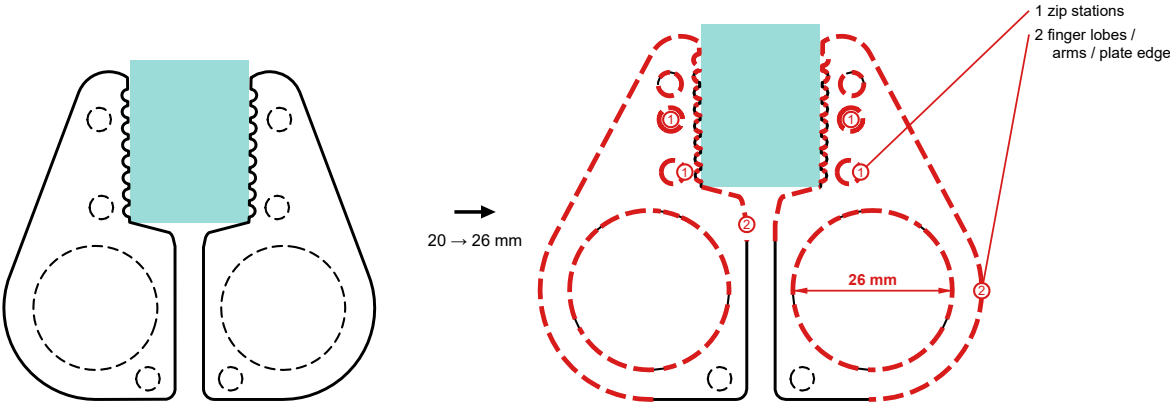
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: hand size set to Large. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Default	Medium
Choices	Small, Medium, Large, Measure my hand
Moves	arms, finger lobes, zip stations

Finger width

Customizer name: **measure_finger_width**

Widens both finger holes and their lobes.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 20 mm wide plug in teal. Right: finger width at 26 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes, the arms and the plate edge: the two rounded lobes in the lower half; the two toothed arms in the upper half; the outer outline.

Default	20 mm
Range	14 to 32 mm, in steps of 0.5 mm
Moves	arms, finger lobes, plate edge, zip stations

Note: Only acts when size is Measure my hand.

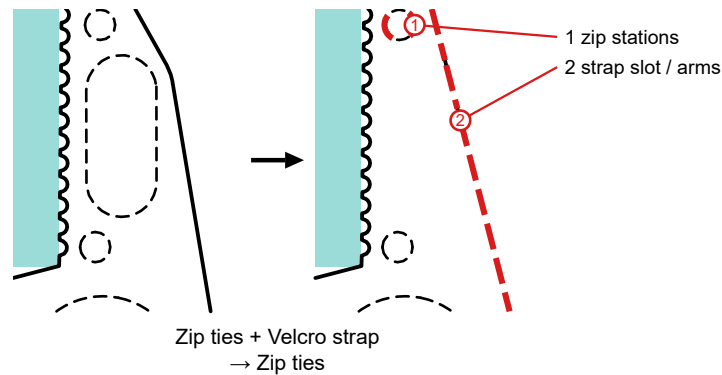
Step 3 - Attachment

Pick how the tool attaches.

Attachment

Customizer name: **attachment**

Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: attachment set to Zip ties. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

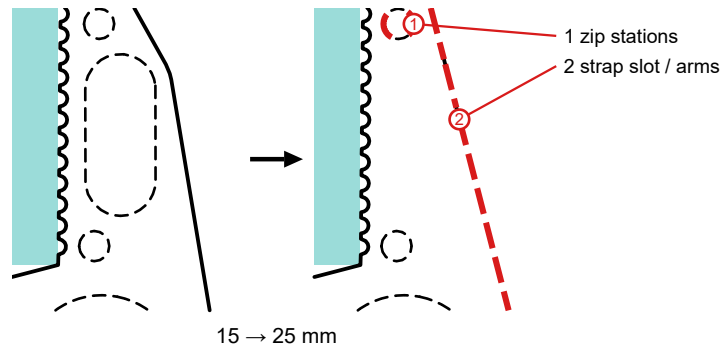
Default	Zip ties + Velcro strap
Choices	Zip ties + Velcro strap, Zip ties
Moves	arms, strap slot, zip stations

Note: Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Strap width

Customizer name: **strap_width**

Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap width at 25 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Default	15 mm
Range	10 to 25 mm, in steps of 1 mm
Moves	arms, strap slot, zip stations
Red warnings	STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP

Note: Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot. At 25 mm the slot leaves too, because the plug is too short for one, so the after picture has no slot and no dimension line.

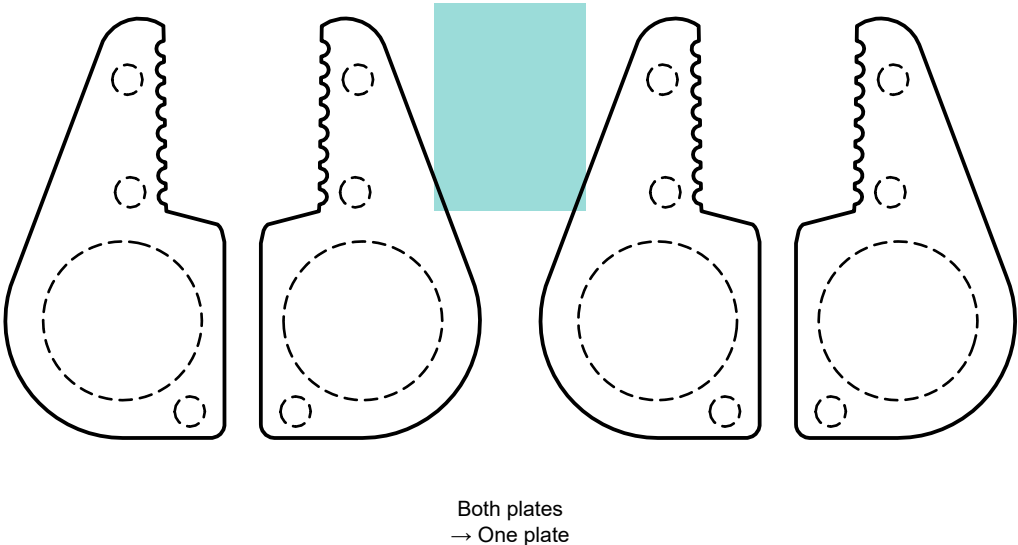
Step 4 - Print Layout

Pick whether the file holds both plates or one.

Print layout

Customizer name: **print_layout**

Puts both identical plates side by side in one file, or just one plate.



One top view of the two-sided puller with both plates side by side, the plug end at the top: what the file prints at Both plates. At One plate it prints one plate. Nothing is marked in red: this dial changes the layout, not the plate.

Default	Both plates
Choices	Both plates, One plate
Moves	arms, finger lobes, plate edge, zip stations

Render, export and print

1. Press **F6** (or **Design ▶ Render**) and wait for the progress bar to finish. In the browser, press **Render**. A render takes from a few seconds to a minute on a computer, longer in a browser.
2. Look for red text beside the part. If there is any, read it: it names the measurement to fix, and the part will not fit until it is gone.
3. Save the file: **File ▶ Export ▶ Export as STL** on the computer, or the download button in the browser. If you export without rendering, OpenSCAD asks you to render first: press F6 and export again. The quick preview (F5) looks complete but cannot be exported.

Load the STL into your slicer with these settings:

Setting	Value
Orientation	both plates flat on the bed, exactly as the file lays them out. No supports.
Layer height	0.2 mm
Walls	3 to 4: you pull hard on this part
Infill	25 to 35 percent
Material	PETG. PLA, ABS and ASA work too. Never a flexible filament: the tool has to stay stiff.

The file holds both plates side by side. Most slicers load it as one object with two parts, which prints fine as it is; use your slicer's split-to-objects command if you want to move one plate on its own.

Assemble

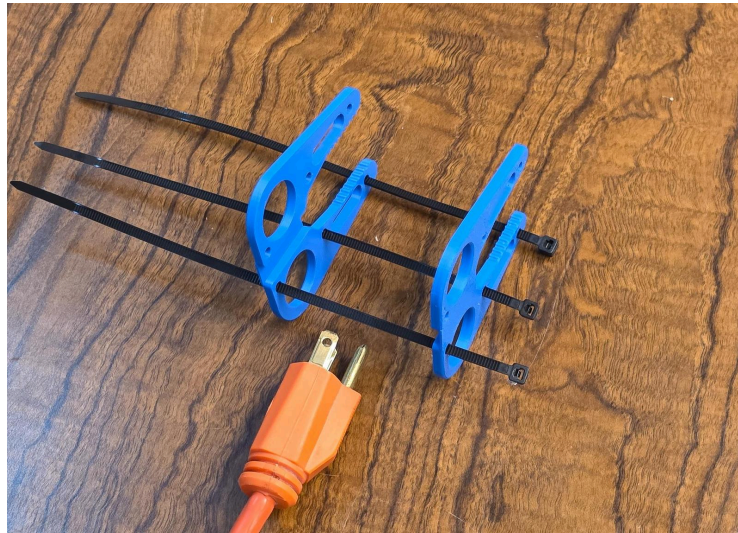
You need three zip ties; the size is in the bill of materials. The two plates are identical; one is flipped over so the toothed arms face each other.

1. Print both plates. The picture shows the two plates on a table with the plug between them.



The two plates of the two-sided puller on a table with the orange extension-cord plug between them, ready to assemble.

2. Thread one end of each zip tie through the first plate, then through the second, so the ties lie between the plates. Make sure the cord channel of both plates faces the same way. The picture shows three ties through one plate, lying toward the other.



Three zip ties threaded through one plate of the two-sided puller and lying flat toward the second plate, the plug beside them.

3. Place your plug between the two plates, its prongs toward the arms. The picture shows the plug lying on the first plate between the ties.



The orange plug placed on the first plate between the zip ties, its prongs toward the plate's arms.

4. Thread the zip ties through the matching holes of the second plate. The picture shows the second plate over the plug with the ties through it and their tails loose.



The second plate laid over the plug, with the zip ties threaded through its matching holes and their tails hanging loose.

5. Pull the zip ties tight enough to hold the plug between the plates. The picture shows the ties cinched with their tails still on.



The zip ties pulled tight around both plates and the orange plug, their tails still sticking out.

6. Cut off the zip-tie tails. The picture shows the finished tool holding the plug.



The finished two-sided puller with the zip-tie tails cut off, holding the orange plug with its prongs clear.

The tool stays on the plug: the serrated arms do the gripping, and the zip ties hold the two plates together.

Use it

The plug puller is for anyone who cannot grip a plug and pull it out of an outlet: arthritis, low grip strength, tremor, a small hand, one hand. Two finger holes take the pull, so your whole hand does the work instead of a fingertip pinch. The tool touches only the plug's sides and back, never the outlet.

The two plates stay zip-tied around the plug, so there is nothing to seat: the tool is already on. The picture shows a two-sided puller and its plug seated in an outlet, ready to be pulled.



The two-sided puller and its orange extension-cord plug seated in a wall outlet, ready to be pulled, no hand on it yet.

1. Put two fingers through the round holes at the cord end, one in each lobe.
2. Pull straight back, away from the wall. The picture shows the pull.



A gloved hand pulling the two-sided puller and its orange extension-cord plug straight out of a wall outlet, two fingers through the holes.

A two-sided puller on a USB-C laptop tip works the same way, pulling the tip out of the laptop instead of a wall.

Safety

- Nothing goes between the plug face and the outlet cover: the tool grips only the plug's sides and back.
- Pull straight out. Never lever the tool sideways or up and down.
- Do not use the tool on a damaged cord, a cracked plug, or a plug that is warm to the touch.
- Keep your fingers away from the prongs as the plug comes out.
- The tool has no conductive parts, but it is still plastic near electricity: if it cracks, stop using it.

If it does not fit

Put the tool on the plug and pull once. The plug should sit flat between the plates with the cord in its channel, and your two fingers should slide in and out of the holes without catching. If anything is tight, loose or awkward, the full guide names the one dial to change and by how much, decodes every red warning tag, and covers the advanced dials: the two-sided puller's full guide.

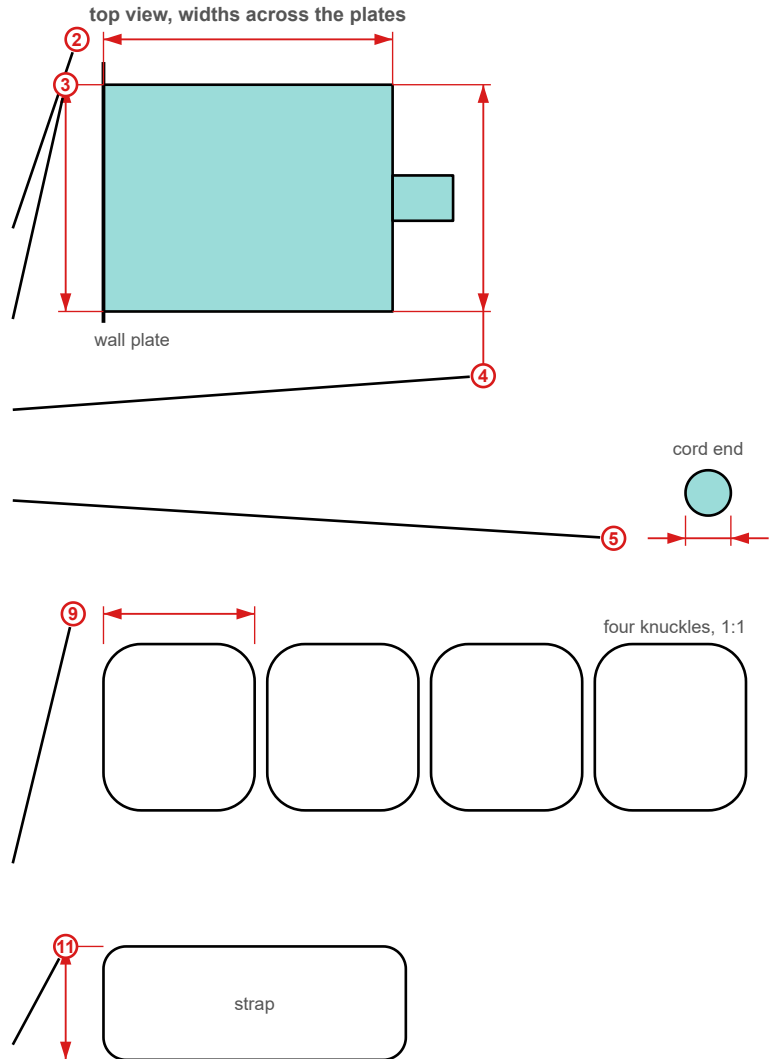
Plug Puller — Measuring form, two-sided puller

Measure in mm. Fill in the blanks top to bottom, then type them into the Customizer in the same order.

Print at 100 %

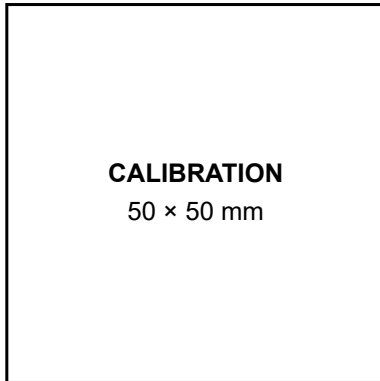
50 mm

- ① plug_preset
Plug preset
 - ☒ **Measure my plug**
 - ☐ Heavy-duty extension cord - NEMA 5-15
 - ☐ USB-C laptop tip ☐ Flat 2-prong lamp plug - NEMA 1-15
 - ☐ Standard 3-prong plug - NEMA 5-15
- ② measure_plug_length
Plug length
default 25.5 mm → _____ mm card R1
- ③ measure_plug_width_prong_end
Plug width at the prong end
default 20 mm → _____ mm card R1
- ④ measure_plug_width_cord_end
Plug width at the cord end
default 20 mm → _____ mm card R1
- ⑤ measure_cord_thickness
Cord thickness
default 4 mm → _____ mm card C1
- ⑥ plug_sides
Plug sides
 - ☒ **Rounded sides** ☐ Flat sides
- ⑦ show_plug_preview
Show the see-through plug
 - ☒ **leave on**
- ⑧ size
Hand size
 - ☐ Small ☒ **Medium** ☐ Large ☐ Measure my hand
- ⑨ measure_finger_width
Finger width
default 20 mm → _____ mm card F1 / F2
- ⑩ attachment
Attachment
 - ☒ **Zip ties + Velcro strap** ☐ Zip ties
- ⑪ strap_width
Strap width
default 15 mm → _____ mm
- ⑫ print_layout
Print layout
 - ☒ **Both plates** ☐ One plate



Plug Puller — Measuring Stencil Sheet

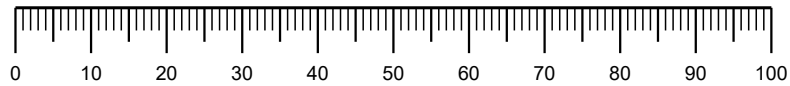
Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.



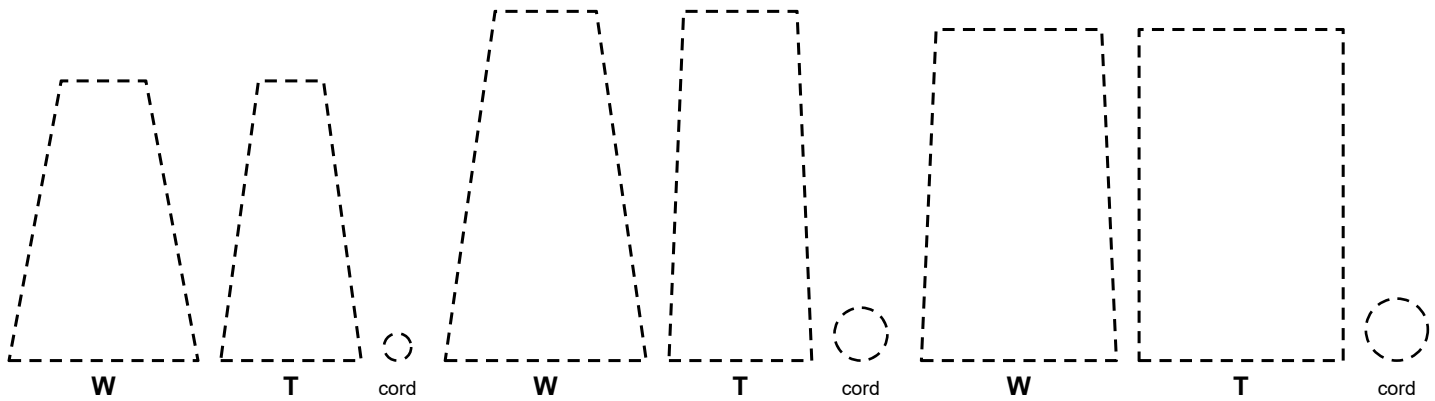
1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your
print was scaled: re-print at 100%.

Card legend: the quick start, under Match a card
Works on A4 and US Letter paper.

R1 — ruler (mm): plug length, widths, thicknesses, hand width



2. Plug silhouettes (1:1) — hold your plug against W (wide side) and T (thin side); if it matches, pick that preset in Step 1.



P1 Lamp 2-prong NEMA 1-15

P2 Standard 3-prong NEMA 5-15

P3 Heavy-duty cord NEMA 5-15

3. F1 / F2 — finger circles: smallest circle your middle finger passes through, minus 5 = finger width.

